Advanced Calculus I Non-hand-in Assignment 3

This homework comes from textbook and will **NOT** be collected (這份作業完成後不需要繳交、評分。). But for practice,or quizzes or exams, please make sure that you have idea to solve the following questions by yourself. (這些題目是很重要的練習，小考、期中、期末考會有類似的題目。練習這些題目的過程中所逐漸形成我們對於所學到的知識的"看法"，會在考試中很有幫助。所以，請一定要了解如何解題與解題背後的想法是什麼。)

Please prove the following propositions are true. (請證明下列命題爲眞。)

Section 8.1.
8.6 (ii) Theorem (on page 288)
8.7 Remark (on page 288)

第 3 週｜自主練習・課本題目補充

William R. Wade | An Introduction to Analysis | Fourth Edition, Global Edition

依教師 Non-hand-in Assignment 指定順序整理；以下為題目／命題敘述，不含解答。

符號補充：下列為 Euclidean norm、1-norm 與 sup-norm。

||x|| = sqrt(sum x_j^2)，||x||_1 = sum |x_j|，||x||_inf = max |x_j|.

Theorem 8.6 (ii)

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8.6 Theorem. *Let* $\mathbf{x}, \mathbf{y} \in \mathbf{R}^n$. *Then*

i) $\|\mathbf{x}\| \geq 0$ *with equality only when* $\mathbf{x} = \mathbf{0}$,
ii) $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$ *for all scalars* α,
iii) [TRIANGLE INEQUALITIES]. $\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ *and* $\|\mathbf{x}-\mathbf{y}\| \geq \|\mathbf{x}\| - \|\mathbf{y}\|$.

本題僅指定 (ii)；此處保留定理完整敘述供辨識假設。

Remark 8.7

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8.7 Remark. *Let* $\mathbf{x} \in \mathbf{R}^n$. *Then*

i) $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\| \leq \sqrt{n}\|\mathbf{x}\|_\infty$, *and*
ii) $\|\mathbf{x}\| \leq \|\mathbf{x}\|_1 \leq \sqrt{n}\|\mathbf{x}\|$.